STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-QC-COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2113-2139 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2141-2158, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_algebraic-stacks-sheaves.json (pb=1025, stmt 2113-2139, proof 2141-2158).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GNE: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2158–2158. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-characterize-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1798–1805. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-characterize-quasi-coherent-bis: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1819–1836. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-restriction: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1185–1200. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DC: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2158–2158. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8H: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2158–2158. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03EL: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2158–2158. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids.*

1. *The category $QCoh(\mathcal{O}_\mathcal{X})$ has colimits and they agree with colimits in the categories $Mod(\mathcal{X}_{Zar}, \mathcal{O}_\mathcal{X})$, $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$, $Mod(\mathcal{O}_\mathcal{X})$, and $LQCoh(\mathcal{O}_\mathcal{X})$.*
2. *Given $\mathcal{F}, \mathcal{G}$ in $QCoh(\mathcal{O}_\mathcal{X})$ the tensor products $\mathcal{F} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{G}$ computed in $Mod(\mathcal{X}_{Zar}, \mathcal{O}_\mathcal{X})$, $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$, or $Mod(\mathcal{O}_\mathcal{X})$ agree and the common value is an object of $QCoh(\mathcal{O}_\mathcal{X})$.*
3. *Given $\mathcal{F}, \mathcal{G}$ in $QCoh(\mathcal{O}_\mathcal{X})$ with $\mathcal{F}$ finite locally free (in fppf, or equivalently étale, or equivalently Zariski topology) the internal homs $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ computed in $Mod(\mathcal{X}_{Zar}, \mathcal{O}_\mathcal{X})$, $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$, or $Mod(\mathcal{O}_\mathcal{X})$ agree and the common value is an object of $QCoh(\mathcal{O}_\mathcal{X})$.*

Proof. Let x be an arbitrary object of $\mathcal{X}$ lying over the scheme U. Let $\tau \in \{Zariski, étale, fppf\}$. To show that an object $\mathcal{H}$ of $Mod(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X})$ is in $QCoh(\mathcal{O}_\mathcal{X})$ it suffices show that the restriction $x^*\mathcal{H}$ (Section section-restriction) is a quasi-coherent object of $Mod((Sch/U)_\tau, \mathcal{O})$. See Lemmas lemma-characterize-quasi-coherent and lemma-characterize-quasi-coherent-bis. Similarly for being finite locally free. Recall that $(Sch/U)_\tau = \mathcal{X}_\tau/x$ is a localization of $\mathcal{X}_\tau$ at an object. Hence restriction commutes with colimits, tensor products, and forming internal hom (see Modules on Sites, Lemmas 03DC, 03EL, and 0E8H). This reduces the lemma to Descent, Lemma 0GNE. □